

ASSIGNMENT 2

Transient (Unsteady) Heat Conduction

Explicit, Implicit, Crank–Nicolson, and ADI Schemes

Maximum Marks: 100

Instructions

1. For *every* problem, implement and compare at least three time-marching schemes: explicit (FTCS), fully implicit (BTCS), and Crank–Nicolson. For the 2D problems (5, 6, 7) also implement the Alternating-Direction-Implicit (ADI) method and include it in the comparison.
2. For each scheme, report: (i) the theoretical and observed stability limit (critical Fourier number) if the scheme is conditionally stable; (ii) accuracy versus the analytical/benchmark solution as a function of time-step size, at fixed grid; and (iii) computational cost (number of time steps $\times$ cost per step, and wall-clock time) to reach a specified dimensionless time.
3. Wherever an analytical or product-solution benchmark is given, validate your code against it quantitatively (tabulated errors), not just by visual comparison of plots.
4. State your convergence/stability criteria, initial conditions, and grid/time-step values clearly in every submission.

Group 1	Group 2	Group 3	Group 4
Problems: 2, 7	Problems: 1, 8	Problems: 4, 5	Problems: 3, 6
230885	230135	230518	231088
230905	230153	230640	231096
251020623	251050046	251290008	252050605
261050013	261050051	261050080	261050611
261050022	261050054	261050086	261050612
261050028	261050058	261050089	261180013
261050032	261050070	261050093	261180018
261050035	261050071	261050402	261180022
261050042	261050072	261050406	261290014
261050045	261050074	261050601	261290016
261050047	261050076	261050605	

Problem 1

Semi-infinite solid with sudden surface temperature change (40 Marks)

Consider a semi-infinite solid (Fig. 1) that is initially at a uniform temperature T_i . At $t = 0$ the surface temperature is suddenly changed to T_∞ . The governing equation (1D Cartesian, no generation) is

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}, \quad x \geq 0, \quad t > 0, \quad T(x, 0) = T_i, \quad T(0, t) = T_\infty \quad (t > 0).$$

Analytical solution (given, with $\theta = T - T_\infty$, $\theta_i = T_i - T_\infty$):

$$\frac{\theta(x, t)}{\theta_i} = 1 - \operatorname{erfc}\left[\frac{x}{2\sqrt{\alpha t}}\right] = \operatorname{erf}\left[\frac{x}{2\sqrt{\alpha t}}\right].$$

Data: $\alpha = 1.0 \times 10^{-5} \text{ m}^2/\text{s}$, $T_i = 20^\circ\text{C}$, $T_\infty = 200^\circ\text{C}$. Truncate the domain at a finite $x_{\max}$ large enough that $T(x_{\max}, t) \approx T_i$ throughout the times of interest (justify your choice).

Tasks:

- Discuss (as originally asked) whether a non-uniform grid would help here, and why — consider how the length scale of the thermal penetration depth $\delta(t) \sim \sqrt{\alpha t}$ changes with time, and how a fixed uniform Δx chosen for early times behaves at later times (and vice versa).
- Discretize using a uniform grid and solve with (i) explicit, (ii) fully implicit, and (iii) Crank–Nicolson schemes. For the explicit scheme, verify the classical 1D stability limit $Fo = \alpha \Delta t / \Delta x^2 \leq \frac{1}{2}$ numerically by running one case just below and one just above this limit.
- Compare all three schemes against the analytical error-function solution at $x = 0.5, 1, 2, 4 \text{ mm}$ for several times, tabulating the maximum error. For a *fixed* Δx , plot error versus Δt for each scheme and confirm the expected $O(\Delta t)$ behaviour for explicit/implicit and $O(\Delta t^2)$ for Crank–Nicolson.
- Repeat with the non-uniform grid you proposed in (a) and quantify the improvement (for the same number of nodes) in accuracy near $x = 0$ at early times.

Problem 2

Plane wall with temperature-dependent thermal conductivity (40 Marks)

A plane wall of thickness L , initially at uniform temperature T_i , is made of a material whose thermal conductivity varies linearly with temperature,

$$k(T) = k_0[1 + \beta(T - T_i)],$$

while density ρ and specific heat c are taken as constant. At $t = 0$ the face $x = 0$ is suddenly raised to T_s and held there; the face $x = L$ is insulated (Fig. 8). Because k depends on the unknown temperature, the field equation must be kept in conservative (divergence) form:

$$\rho c \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left[k(T) \frac{\partial T}{\partial x} \right], \quad 0 \leq x \leq L, \quad T(x, 0) = T_i, \quad T(0, t) = T_s \quad (t > 0), \quad \left. \frac{\partial T}{\partial x} \right|_{x=L} = 0.$$

Data: $L = 0.04 \text{ m}$, $\rho = 7800 \text{ kg/m}^3$, $c = 470 \text{ J/(kg.K)}$, $k_0 = 20 \text{ W/(m.K)}$ (value at $T = T_i$), $\beta = 2.5 \times 10^{-3} \text{ K}^{-1}$, $T_i = 25^\circ\text{C}$, $T_s = 325^\circ\text{C}$.

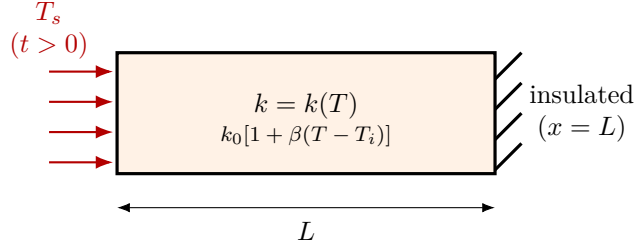


Fig. 8 — Plane wall with temperature-dependent conductivity $k(T)$: face $x = 0$ suddenly raised to T_s , face $x = L$ insulated

Tasks:

- Derive the finite-volume (conservative) discretization for variable $k(T)$: express the flux at each cell face using a suitably averaged conductivity between adjacent nodes (state whether you use the arithmetic or the harmonic mean, and why the harmonic mean is generally preferred when conductivity varies strongly), and write out the resulting explicit (FTCS) update formula.
- For the fully implicit and Crank–Nicolson schemes, the face conductivities depend on the unknown T^{n+1} . Describe and implement a Picard (successive-substitution) iteration at each time step: guess T^{n+1} (e.g. from the previous time level), evaluate k at the guessed values, solve the resulting linear tridiagonal system, update the guess, and repeat to a stated tolerance (e.g. $\max |T^{(m+1)} - T^{(m)}| < 10^{-3} \text{ } ^\circ\text{C}$). Separately, show that the explicit scheme’s stability limit must be based on the *largest* diffusivity $\alpha(T) = k(T)/(\rho c)$ expected in the domain (i.e. at $T = T_s$, not at T_i), and verify this numerically.
- Solve with explicit, fully implicit, and Crank–Nicolson schemes. Plot $T(x, t)$ at several times and the surface heat flux history $q''(0, t) = -k(T_0) \partial T / \partial x|_{x=0}$. As a built-in check, set $\beta = 0$ and confirm your code collapses to the constant-property plane-wall problem (you may validate this limiting case against the plane-wall series already used in Problem 4).
- (*Bonus*) Apply the Kirchhoff transformation $u = \frac{1}{k_0} \int_{T_i}^T k(T') dT'$ to recast the field equation approximately as a linear diffusion equation in u (state clearly what assumption on ρc this requires), solve for $u(x, t)$ with the transformed boundary conditions, back-transform to recover $T(x, t)$, and compare with your direct nonlinear solution from (c).
- Compare the three schemes on cost (including the number of Picard iterations per step needed by the implicit and Crank–Nicolson schemes) and accuracy, and discuss whether Crank–Nicolson’s higher time-accuracy is still worth its extra cost here, given that iteration is required at every step regardless of scheme.

Problem 3

Convective quench of a solid cylinder (1D, cylindrical coordinates) (40 Marks)

A long solid cylinder of radius R , initially at uniform temperature T_i , is suddenly exposed at $t = 0$ to a convective environment (h, T_∞) over its entire curved surface (Fig. 3). Find $T(r, t)$.

$$\frac{\partial T}{\partial t} = \alpha \left[\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} \right], \quad 0 \leq r \leq R, \quad \frac{\partial T}{\partial r} \Big|_{r=0} = 0, \quad -k \frac{\partial T}{\partial r} \Big|_{r=R} = h(T(R, t) - T_\infty).$$

Data: $R = 0.02$ m, $\alpha = 1.0 \times 10^{-5}$ m²/s, $k = 20$ W/(m.K), $h = 200$ W/(m².K) (so $Bi = hR/k = 0.2$), $T_i = 300^\circ\text{C}$, $T_\infty = 25^\circ\text{C}$.

Analytical solution (series; $\theta = T - T_\infty$, $\theta_i = T_i - T_\infty$, $Fo = \alpha t/R^2$):

$$\frac{\theta(r,t)}{\theta_i} = \sum_{n=1}^{\infty} C_n e^{-\lambda_n^2 Fo} J_0\left(\lambda_n \frac{r}{R}\right), \quad C_n = \frac{2}{\lambda_n} \frac{J_1(\lambda_n)}{J_0^2(\lambda_n) + J_1^2(\lambda_n)},$$

where λ_n are the positive roots of $\lambda_n J_1(\lambda_n) = Bi J_0(\lambda_n)$.

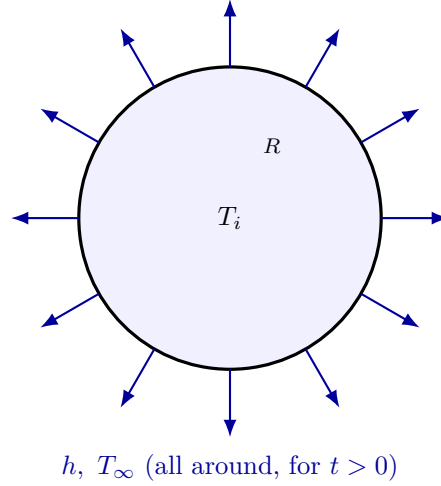


Fig. 3 — Solid cylinder suddenly convected over its entire curved surface

Tasks:

- Show that at $r = 0$, using symmetry and l'Hôpital's rule, the governing equation reduces to $\partial T/\partial t = 2\alpha \partial^2 T/\partial r^2$, and derive the explicit finite-difference update at the centre node. Hence show that the explicit scheme is stable there only for $Fo = \alpha \Delta t/\Delta r^2 \leq \frac{1}{4}$ — *twice as restrictive* as the ordinary interior-node limit $Fo \leq \frac{1}{2}$, and therefore governs the whole domain.
- Discretize the outer convective boundary and assemble the full system. Solve with explicit, fully implicit, and Crank–Nicolson schemes.
- Find the first few roots λ_n of the transcendental equation above (numerically), evaluate the series (enough terms for 5-figure convergence), and compare your three numerical schemes against it at $r = 0, R/2, R$ for $Fo = 0.1, 0.3, 0.6$. Tabulate the maximum error for each scheme/time step.
- Compare the number of time steps and CPU time required by each scheme for a fixed accuracy target, and recommend a scheme for this problem.

Problem 4

Transient conduction in a thin-walled hollow cylinder (40 Marks)

A hollow cylinder (pipe wall), $R_i \leq r \leq R_o$, is initially at uniform temperature T_i . At $t = 0$ the inner surface is suddenly raised to T_s and held there; the outer surface is insulated (Fig. 4).

$$\frac{\partial T}{\partial t} = \alpha \left[\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} \right], \quad R_i \leq r \leq R_o, \quad T(R_i, t) = T_s \ (t > 0), \quad \left. \frac{\partial T}{\partial r} \right|_{r=R_o} = 0.$$

Data: $R_i = 0.045$ m, $R_o = 0.05$ m (thin wall, thickness $\Delta = R_o - R_i = 5$ mm), $\alpha = 1.2 \times 10^{-5}$ m²/s, $T_i = 20^\circ\text{C}$, $T_s = 300^\circ\text{C}$.

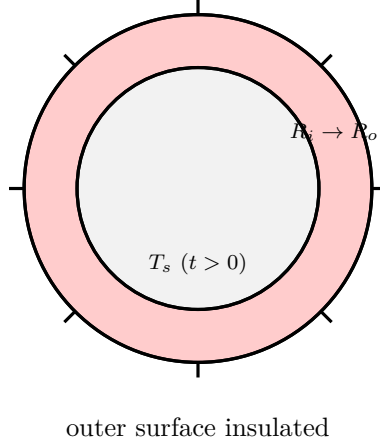


Fig. 4 — Thin hollow cylinder: inner surface suddenly raised to T_s , outer surface insulated (hatched)

No simple closed form exists for this annulus problem (the eigenfunctions mix Bessel functions J_0, Y_0). Instead, validate your code using two checks:

- **Long-time limit:** since the outer surface is insulated and there is no generation, $T(r, t) \rightarrow T_s$ everywhere as $t \rightarrow \infty$ (no place for heat to go). Confirm this numerically.
- **Thin-wall (planar) limit:** because $\Delta/R_i = 0.11 \ll 1$ here, curvature effects are small, and the problem should closely match the classical 1D *plane wall* solution (one face suddenly at T_s , the other insulated), with x measured from the insulated face:

$$\frac{\theta(x, t)}{\theta_i} = \sum_{n=0}^{\infty} \frac{4(-1)^n}{(2n+1)\pi} \exp\left[-\frac{(2n+1)^2\pi^2}{4} Fo\right] \cos\left(\frac{(2n+1)\pi x}{2\Delta}\right), \quad Fo = \frac{\alpha t}{\Delta^2},$$

with $\theta = T - T_s$, $\theta_i = T_i - T_s$.

Tasks:

- Discretize the annulus (regular interior cylindrical formula; insulated outer BC via a ghost node) and solve with explicit, implicit, and Crank–Nicolson schemes.
- Verify the long-time limit and, separately, compare your transient solution against the plane-wall series above (mapping $x = r - R_i$) at a few times. Quantify the discrepancy caused by curvature and comment on whether it grows or shrinks as Fo increases.

- (c) Repeat for a *thick* annulus, $R_i = 0.02$ m, $R_o = 0.05$ m (same α, T_i, T_s), where the planar approximation is no longer expected to hold. Quantify how much worse the planar comparison becomes, confirming that curvature effects scale with Δ/R_i .
- (d) Compare the three time-marching schemes as in previous problems (stability limit, accuracy vs. Δt , cost).

Problem 5

Square bar suddenly placed in a radiating (black-body) furnace (60 Marks)

An infinitely long bar of thermal diffusivity α has a square cross-section of side $2L$ (Fig. 2). It is initially at a uniform temperature T_i when it is placed quickly inside a furnace that behaves as a black-body enclosure at T_b . Using the non-dimensional variables

$$X = \frac{x}{L}, \quad Y = \frac{y}{L}, \quad \tau = \frac{\alpha t}{L^2}, \quad \theta = \frac{T - T_i}{T_b - T_i}, \quad \beta = \frac{k}{\varepsilon \sigma L T_b^3}, \quad \gamma = \frac{T_i}{T_b},$$

write the governing differential equation, and the initial and boundary conditions (the outer-surface BC is the nonlinear radiative balance $-k \partial T / \partial n = \varepsilon \sigma (T^4 - T_b^4)$), and show that the general solution has the form $\theta = \theta(X, Y, \tau, \beta, \gamma)$.

Suggested test data: steel bar, $L = 0.3$ m, $\alpha = 1.16 \times 10^{-5}$ m²/s, $T_i = 300$ K, $k = 45$ W/(m.K), $\varepsilon = 0.79$, furnace $T_b = 1273$ K.

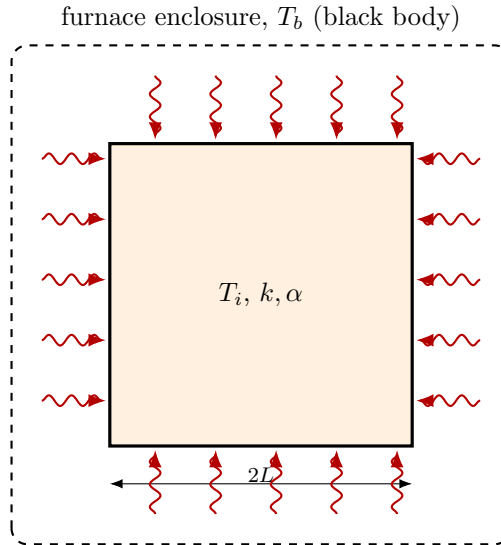


Fig. 2 — Square bar (side $2L$) suddenly placed inside a black-body radiating enclosure at T_b

Tasks:

- (a) Complete the non-dimensionalization requested above, clearly showing where β and γ enter the boundary condition.
- (b) Linearize the radiative boundary condition at each time step (e.g. $\varepsilon \sigma (T^4 - T_b^4) \approx h_r (T - T_b)$ with $h_r = \varepsilon \sigma (T + T_b)(T^2 + T_b^2)$ evaluated at the previous iterate/time level), and discretize the 2D problem in space (5-point Laplacian, using symmetry to model one quadrant).

- (c) Solve using explicit, fully implicit, and Crank–Nicolson time marching (the latter two require an inner iteration per step because h_r depends on the unknown surface temperature). Find the time for the centre temperature to reach 786.5 K ($\theta_c = 0.5$) for $L = 0.3$ m and for $L = 0.6$ m, as originally asked.
- (d) Compare the number of time steps and the wall-clock time needed by each scheme to reach $\theta_c = 0.5$ within a fixed tolerance. Discuss why the explicit scheme's *effective* stability limit here is not a fixed Fourier number, but tightens as the surface heats up (since the linearized radiative coefficient h_r grows with T^3).

Problem 6

Convective quench of a long rectangular bar (2D Cartesian, product solution) (60 Marks)

A long bar of rectangular cross-section $a \times b$, initially at uniform T_i , is suddenly exposed at $t = 0$ to convection (h, T_∞) on all four side faces (Fig. 5).

$$\frac{\partial T}{\partial t} = \alpha \left[\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right], \quad |x| \leq a/2, \quad |y| \leq b/2,$$

with convective BCs on all four faces and symmetry ($\partial T/\partial x = \partial T/\partial y = 0$) at the centreline $x = 0, y = 0$.

Data: $a = 0.10$ m, $b = 0.06$ m, $k = 15$ W/(m.K), $\alpha = 4.0 \times 10^{-6}$ m²/s, $h = 100$ W/(m².K), $T_i = 200^\circ\text{C}$, $T_\infty = 25^\circ\text{C}$. (Half-widths $L_1 = a/2 = 0.05$ m, $L_2 = b/2 = 0.03$ m; $Bi_1 = hL_1/k = 0.333$, $Bi_2 = hL_2/k = 0.2$.)

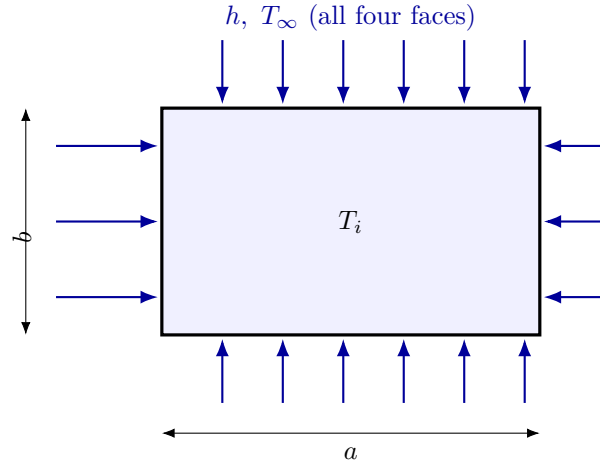


Fig. 5 — Rectangular bar suddenly exposed to convection on all four faces

Analytical (product-solution) benchmark. With x, y measured from the centreline, the 1D convectively-cooled infinite-slab solution (half-thickness L , $Bi = hL/k$, $Fo = \alpha t/L^2$) is

$$\frac{\theta_{\text{slab}}(x, t)}{\theta_i} = \sum_{n=1}^{\infty} \frac{4 \sin \lambda_n}{2\lambda_n + \sin 2\lambda_n} e^{-\lambda_n^2 Fo} \cos\left(\lambda_n \frac{x}{L}\right), \quad \lambda_n \tan \lambda_n = Bi.$$

The rectangular bar's 2D solution is the *product* of two such 1D solutions, one per direction:

$$\frac{\theta(x, y, t)}{\theta_i} = \frac{\theta_{\text{slab},1}(x, t)}{\theta_i} \cdot \frac{\theta_{\text{slab},2}(y, t)}{\theta_i}.$$

Tasks:

- Derive the 2D explicit finite-difference update and show that, for $\Delta x = \Delta y = h$, stability requires $Fo = \alpha \Delta t / h^2 \leq \frac{1}{4}$ (i.e. half as permissive as the 1D limit, because both directions contribute to diffusive loss from each node).
- Implement explicit, fully implicit, Crank–Nicolson, *and* ADI (Peaceman–Rachford) time marching. For ADI, state the tridiagonal sweep sequence you use.
- Compute the two 1D slab solutions (find λ_n roots for Bi_1 and Bi_2 separately) and form the product-solution benchmark. Compare your 2D numerical results against it at the centre and at $(a/4, 0)$, $(0, b/4)$ for several Fo , tabulating the maximum error.
- Compare all four schemes on cost (time steps, CPU time, and — for ADI — number of tridiagonal solves per step) needed to reach a fixed accuracy, and discuss why ADI is attractive for 2D problems relative to a fully implicit scheme solved with a general (non-ADI) linear solver.

Problem 7

Convective quench of a short cylinder (2D, cylindrical r – z , product solution) (60 Marks)

A short solid cylinder of radius R and half-height L_z (total height $2L_z$), initially at uniform T_i , is suddenly exposed at $t = 0$ to convection (h, T_∞) on *both* the curved lateral surface and the two flat end faces (Fig. 6). This problem combines the radial physics of Problem 3 with the axial physics of Problem 5.

$$\frac{\partial T}{\partial t} = \alpha \left[\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} + \frac{\partial^2 T}{\partial z^2} \right], \quad 0 \leq r \leq R, \quad |z| \leq L_z.$$

Data: $R = 0.04$ m, $L_z = 0.03$ m, $\alpha = 1.0 \times 10^{-5}$ m²/s, $k = 20$ W/(m.K), $h = 150$ W/(m².K), $T_i = 250^\circ\text{C}$, $T_\infty = 25^\circ\text{C}$. ($Bi_R = hR/k = 0.3$, $Bi_z = hL_z/k = 0.225$.)

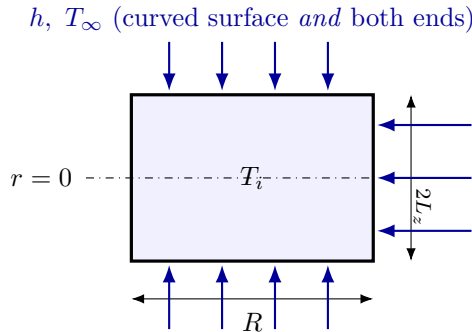


Fig. 6 — Axisymmetric (r – z) section of a short cylinder, convected on the curved lateral surface and both end faces

Analytical (product-solution) benchmark. The short cylinder's solution is the product of Problem 3's infinite-cylinder solution (radius R , giving $\theta_{\text{cyl}}(r, t)/\theta_i$) and Problem 5's infinite-slab solution (half-thickness L_z , giving $\theta_{\text{slab}}(z, t)/\theta_i$):

$$\frac{\theta(r, z, t)}{\theta_i} = \frac{\theta_{\text{cyl}}(r, t)}{\theta_i} \cdot \frac{\theta_{\text{slab}}(z, t)}{\theta_i}.$$

Tasks:

- Combine the axis treatment of Problem 3(a) (for $r = 0$) with the standard Cartesian treatment in z to discretize the full r - z domain. State the combined explicit stability limit (radial and axial Fourier numbers both must satisfy their respective single-direction limits simultaneously; identify which one governs for your chosen $\Delta r, \Delta z$).
- Implement explicit, fully implicit, Crank–Nicolson, and ADI schemes (alternating sweeps in r and z).
- Build the product-solution benchmark by re-using (do not re-derive) your Problem 3 cylinder series (with Bi_R) and Problem 5 slab series (with Bi_z). Compare your 2D (r, z) numerical solution against it at the geometric centre, on the mid-plane at $r = R/2$, and on the axis at $z = L_z/2$, for several $For = \alpha t/R^2$.
- Compare the four schemes on cost and accuracy as before, and comment on whether the more restrictive of the two 1D stability limits (radial vs. axial) controls the allowable Δt for the explicit scheme in 2D.

Problem 8

Solid cylinder with asymmetric convective cooling (2D, cylindrical r - θ) (60 Marks)

A long solid cylinder of radius R , initially at uniform T_i , is suddenly quenched at $t = 0$ with *different* convective conditions on the two halves of its circumference (Fig. 7): a strong quench ($h_1, T_{\infty 1}$) on $0 < \theta < \pi$ and a milder one ($h_2, T_{\infty 2}$) on $\pi < \theta < 2\pi$.

$$\frac{\partial T}{\partial t} = \alpha \left[\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} \right], \quad 0 \leq r \leq R, \quad 0 \leq \theta < 2\pi,$$

with T bounded and smooth at $r = 0$, periodic in θ , and

$$-k \frac{\partial T}{\partial r} \Big|_R = \begin{cases} h_1 (T(R, \theta, t) - T_{\infty 1}), & 0 < \theta < \pi \\ h_2 (T(R, \theta, t) - T_{\infty 2}), & \pi < \theta < 2\pi. \end{cases}$$

Data: $R = 0.025$ m, $\alpha = 1.0 \times 10^{-5}$ m²/s, $k = 20$ W/(m.K), $T_i = 300^\circ\text{C}$; upper half $h_1 = 300$ W/(m².K), $T_{\infty 1} = 20^\circ\text{C}$; lower half $h_2 = 50$ W/(m².K), $T_{\infty 2} = 20^\circ\text{C}$.

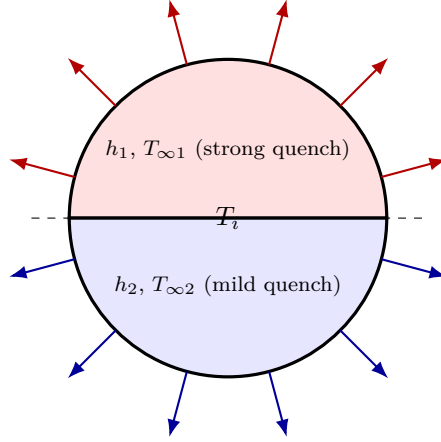


Fig. 7 — Solid cylinder quenched with two different convective conditions on its two circumferential halves

No closed-form solution is expected (the discontinuous, θ -dependent boundary condition breaks simple product separability). Validate instead by a **built-in limiting case**: if you set $h_1 = h_2$ and $T_{\infty 1} = T_{\infty 2}$, your 2D (r, θ) code must reproduce the Problem 3 axisymmetric (1D) Bessel-series solution exactly, since the problem is then axisymmetric.

Tasks:

- (a) Discretize the domain combining the pole treatment of Problem 3(a) with periodicity in θ (as in earlier assignments). Implement explicit, fully implicit, and Crank–Nicolson schemes (ADI, alternating r - and θ -sweeps, is optional/bonus here).
- (b) **Required validation:** run your code with $h_1 = h_2 = 200 \text{ W}/(\text{m}^2\cdot\text{K})$, $T_{\infty 1} = T_{\infty 2} = 20^\circ\text{C}$ and confirm it reproduces the Problem 3 solution (same R, α, k , appropriately matched Bi) to within your discretization error, at several (r, Fo) combinations.
- (c) Run the true asymmetric case with the data above. Plot $T(R, \theta, t)$ at several times and identify how long it takes for the hot and cold sides of the surface to differ by less than 5°C of their eventual (steady-state) difference.
- (d) Since the boundary condition remains permanently asymmetric, explain why the *steady-state* temperature field itself must also be asymmetric, and describe (without necessarily solving it) how you would obtain that steady-state field as $t \rightarrow \infty$ in your code, and how you would confirm your transient solution is indeed approaching it.
- (e) Compare the three schemes (stability, accuracy relative to the validation case in (b), and cost) as in the previous problems.